

Task 53: Find K Pairs with Smallest Sums

Problem Description

You are given two integer arrays `nums1` and `nums2` sorted in non-decreasing order and an integer `k`. Define a pair (u, v) which consists of one element from the first array and one element from the second array. Return the `k` pairs $(u_1, v_1), (u_2, v_2), \dots, (u_k, v_k)$ with the smallest sums.

Java Solution

```
class Solution {
    public List<List<Integer>> kSmallestPairs(int[] nums1, int[] nums2,
int k) {
        PriorityQueue<int[]> pq = new PriorityQueue<>((a, b) -> a[0] +
a[1] - b[0] - b[1]);
        List<List<Integer>> res = new ArrayList<>();
        if (nums1.length == 0 || nums2.length == 0 || k == 0) return res;

        for (int i = 0; i < nums1.length && i < k; i++) {
            pq.offer(new int[]{nums1[i], nums2[0], 0});
        }

        while (k-- > 0 && !pq.isEmpty()) {
            int[] cur = pq.poll();
            res.add(Arrays.asList(cur[0], cur[1]));
            if (cur[2] == nums2.length - 1) continue;
            pq.offer(new int[]{cur[0], nums2[cur[2] + 1], cur[2] + 1});
        }

        return res;
    }
}
```